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SmartSense - API Documentation

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G705

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1. Introduction

The Maintenance Management System API provides a complete RESTful interface for managing industrial maintenance operations, including:

- Users (administrators, directors, technicians)
- Machines
- Problems reported by users or sensors
- Maintenance tasks
- Assistance requests
- Sensor readings

The API follows REST principles, uses JSON for all data, and relies on JWT authentication for secure access.

Base URL:

`http://localhost:8080/api/v1`

All endpoints return a JSON response.

<http://localhost:8080/> to show that API is Online.

2. Auth Controller

2.1 POST

POST api/v1/auth/register

Description:

Register a user and gives a JWT token.

Request Body:

```
{
  "id": 0,
  "name": "string",
  "email": "string",
  "role": "ADMIN",
  "active": true,
  "online": true,
  "privileged": true,
  "password": "string"
}
```

Successful Response (200):

```
{}
```

Error Responses:

```
Error: response status is 500
```

Usage:

Include the token in all protected endpoints:

Authorization: Bearer <token>

POST api/v1/auth/login

Description:

Authenticates a user and returns a JWT token.

Request Body:

```
{
  "email": "user@example.com",
  "password": "1234"
}
```

Successful Response (200):

```
{
  "token": "string",
  "role": "string",
  "name": "string",
  "id": 0
}
```

Error Responses:

```
401 Unauthorized – Invalid credentials
```

Usage:

Include the token in all protected endpoints:

Authorization: Bearer <token>

3. User Controller

3.1 POST

POST /api/v1/users

Creates a new user.

Request Body:

```
{
  "name": "user2",
  "email": "user2@example.com",
  "role": "TECHNICIAN",
  "active": true,
  "online": true,
  "privileged": true,
  "password": "1234",
}
```

Response 201:

```
[{ ...created user... }]
```

3.2 GET

GET /api/v1/users

Returns all non-archived users.

Response 200:

```
[
  {
    "id": 0,
    "name": "user",
    "email": "user@example.com",
    "role": "ADMIN",
    "active": true,
    "online": true,
    "privileged": true,
    "password": "string"
  }
]
```

GET /api/v1/users{id}

Returns a user by ID.

Response 200:

```
[{ ...user object... }]
```

Error

```
404 User not found
```

3.3 PUT

PUT /api/v1/users/{id}

Description:

Updates an existing user.

Request Body:

```
{
  "id": 0,
  "name": "string",
  "email": "string",
  "role": "ADMIN",
  "active": true,
  "online": true,
  "privileged": true,
  "password": "string"
}
```

Response 200:

```
{ ...updated user... }
```

PUT /api/v1/users/{id}/archive

Description:

Archives a user by ID (soft delete).

Response 200:

```
{
  "id": 0,
  "name": "string",
  "email": "string",
  "role": "ADMIN",
  "active": true,
  "online": true,
  "privileged": true,
  "password": "string"
}
```

PUT /api/v1/users/{id}/restore

Description:

Restores an archived user by ID.

Response 200:

```
{
  "id": 0,
  "name": "string",
  "email": "string",
  "role": "ADMIN",
  "active": true,
  "online": true,
  "privileged": true,
  "password": "string"
}
```

3.4 Delete

DELETE /api/v1/users/{id}

Description:

Permanently deletes a user.

Response:

204 No Content

4. Technicians API

4.1 GET

GET /api/v1/users/technicians

Description:

Returns all non-archived technicians.

Response 200:

```
[
  {
    "id": 0,
    "name": "string"
  }
]
```

GET /api/v1/users/technicians/archived

Description:

Returns all archived technicians.

Response 200:

```
[
  {
    "id": 0,
    "name": "string"
  }
]
```

5. Directors API

5.1 GET

GET /api/v1/users/directors

Description:

Returns all directors.

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "email": "string",
    "role": "DIRECTOR",
    "active": true,
    "online": true,
    "privileged": true,
    "password": "string"
  }
]
```

6. Machine Management API

6.1 POST

POST /api/v1/machines

Description:

Create a machine

```
{
  "id": 0,
  "name": "string",
  "location": "string",
  "importanceLevel": 0,
  "status": "ACTIVE",
  "downtimeSum": 0,
  "suspicionFlag": true,
  "vibrationSensor": true,
  "temperatureSensor": true,
  "pressureSensor": true,
  "actionRequiredCount": 0,
  "assistanceRequestedCount": 0,
  "assignedTechnicians": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```

Response 200:

```
{ ...machine data... }
```

6.2 GET

GET /api/v1/machines

Description:

Return all machines

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "location": "string",
    "importanceLevel": 0,
    "status": "ACTIVE",
    "downtimeSum": 0,
    "suspicionFlag": true,
    "vibrationSensor": true,
    "temperatureSensor": true,
    "pressureSensor": true,
    "actionRequiredCount": 0,
    "assistanceRequestedCount": 0,
    "assignedTechnicians": [
      {
        "id": 0,
        "name": "string"
      }
    ]
  }
]
```

GET /api/v1/machines/{id}

Description:

Return machines by ID.

Response 200:

```
{
  "id": 0,
  "name": "string",
  "location": "string",
  "importanceLevel": 0,
  "status": "ACTIVE",
  "downtimeSum": 0,
  "suspicionFlag": true,
  "vibrationSensor": true,
  "temperatureSensor": true,
  "pressureSensor": true,
  "actionRequiredCount": 0,
  "assistanceRequestedCount": 0,
  "assignedTechnicians": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```

GET /api/v1/machines/ranking

Description:

Return machines by ranking order.

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "location": "string",
    "priority": 0
  }
]
```

GET /api/v1/machines/active

Description:

Return all active machines.

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "location": "string",
    "importanceLevel": 0,
    "status": "ACTIVE",
    "downtimeSum": 0,
    "suspicionFlag": true,
    "vibrationSensor": true,
    "temperatureSensor": true,
    "pressureSensor": true,
    "actionRequiredCount": 0,
    "assistanceRequestedCount": 0,
    "assignedTechnicians": [
      {
        "id": 0,
        "name": "string"
      }
    ]
  }
]
```

GET /api/v1/machines/archived

Description:

Return all archived machines.

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "location": "string",
    "importanceLevel": 0,
    "status": "ARCHIVED",
    "downtimeSum": 0,
    "suspicionFlag": true,
    "vibrationSensor": true,
    "temperatureSensor": true,
    "pressureSensor": true,
    "actionRequiredCount": 0,
    "assistanceRequestedCount": 0,
    "assignedTechnicians": [
      {
        "id": 0,
        "name": "string"
      }
    ]
  }
]
```

GET /api/v1/machines/assigned/{technicianID}

Description:

Return all machines assigned to the technicians.

Response 200:

```
[
  {
    "id": 0,
    "name": "string",
    "location": "string",
    "importanceLevel": 0,
    "status": "ARCHIVED",
    "downtimeSum": 0,
    "suspicionFlag": true,
    "vibrationSensor": true,
    "temperatureSensor": true,
    "pressureSensor": true,
    "actionRequiredCount": 0,
    "assistanceRequestedCount": 0,
    "assignedTechnicians": [
      {
        "id": 0,
        "name": "string"
      }
    ]
  }
]
```

6.3 PUT

PUT /api/v1/machines/{id}

Description:

Update machine by ID.

```
{
  "id": 0,
  "name": "string",
  "location": "string",
  "importanceLevel": 0,
  "status": "ACTIVE",
  "downtimeSum": 0,
  "suspicionFlag": true,
  "vibrationSensor": true,
  "temperatureSensor": true,
  "pressureSensor": true,
  "actionRequiredCount": 0,
  "assistanceRequestedCount": 0,
  "assignedTechnicians": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```

Response 200:

```
{ ...machine updated...
}
```

6.4 PATCH

PATCH /api/v1/users/{id}/archive

Description:

Archives a machine by ID (soft delete).

Response 200:

```
{
  "id": 0,
  "name": "string",
  "location": "string",
  "importanceLevel": 0,
  "status": "ARCHIVE",
  "downtimeSum": 0,
  "suspicionFlag": true,
  "vibrationSensor": true,
  "temperatureSensor": true,
  "pressureSensor": true,
  "actionRequiredCount": 0,
  "assistanceRequestedCount": 0,
  "assignedTechnicians": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```

PATCH /api/v1/users/{id}/restore

Description:

Restores a machine by ID.

Response 200:

```
{
  "id": 0,
  "name": "string",
  "location": "string",
  "importanceLevel": 0,
  "status": "ACTIVE",
  "downtimeSum": 0,
  "suspicionFlag": true,
  "vibrationSensor": true,
  "temperatureSensor": true,
  "pressureSensor": true,
  "actionRequiredCount": 0,
  "assistanceRequestedCount": 0,
  "assignedTechnicians": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```

6.5 Delete

DELETE /api/v1/machines/{id}

Description:

Permanently deletes a user.

Response:

204 OK

7. Problem Controller

7.1 POST

POST /api/v1/problems

Description:

Create a problem.

```
{
  "id": 0,
  "machineId": 0,
  "description": "string",
  "resolved": false,
  "startProblemDate": "string",
  "solvedProblemDate": "string"
}
```

Response 204:

```
{
  "id": 0,
  "machineId": 0,
  "description": "string",
  "resolved": false,
  "startProblemDate": "string",
  "solvedProblemDate": "string"
}
```

7.2 GET

GET /api/v1/problems

Description:

Return all machine problems

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "description": "string",
    "resolved": false,
    "startProblemDate": "string",
    "solvedProblemDate": "string"
  }
]
```

GET /api/v1/problems

Description:

Return all machine problems

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "description": "string",
    "resolved": false,
    "startProblemDate": "string",
    "solvedProblemDate": "string"
  }
]
```

GET /api/v1/problems/{id}

Description:

Return the machine problems by id.

Response 204:

```
{
  "id": 0,
  "machineId": 0,
  "description": "string",
  "resolved": false,
  "startProblemDate": "string",
  "solvedProblemDate": "string"
}
```

GET /api/v1/problems/machine/{machineId}

Description:

Return the machine problems by id machine.

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "description": "string",
    "resolved": false,
    "startProblemDate": "string",
    "solvedProblemDate": "string"
  }
]
```

GET /api/v1/problems/history

Description:

Return the history problem

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "description": "string",
    "resolved": false,
    "startProblemDate": "string",
    "solvedProblemDate": "string"
  }
]
```

7.3 PUT

PUT /api/v1/problems/{id}

Description:

Update the problem by id.

```
{
  "id": 0,
  "machineId": 0,
  "description": "string",
  "resolved": false,
  "startProblemDate": "string",
  "solvedProblemDate": "string"
}
```

Response 204:

```
{ ...problem updated...
}
```

7.4 PATCH

PATCH /api/v1/problems/{id}/resolve

Description:

Update the problem by id to resolved.

Response 204:

```
{
  "id": 0,
  "machineId": 0,
  "description": "string",
  "resolved": true,
  "startProblemDate": "string",
  "solvedProblemDate": "string"
}
```

7.5 Delete

DELETE /api/v1/problems/{id}

Description:

Permanently deletes a problem.

Response:

204 OK

8. Maintenance Controller

8.1 POST

POST /api/v1/maintenances

Description:

Create a maintenance request.

```
{
  "id": 0,
  "machineId": 0,
  "technicianId": 0,
  "type": "NORMAL",
  "status": "PENDING",
  "notes": "string"
}
```

Response 204:

```
{... maintenance data...}
```

POST /api/v1/maintenances

Description:

Create a maintenance request in start phase.

```
{
  "id": 0,
  "technicianId": 0,
  "technicianName": "string",
  "machineId": 0,
  "machineName": "string",
  "startTime": "2026-05-13T19:08:37.712Z",
  "endTime": "2026-05-13T19:08:37.712Z",
  "active": true
}
```

Response 204:

```
{... maintenance data...}
```

8.2 GET

GET /api/v1/maintenance

Description:

Return all maintenance requests.

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "technicianId": 0,
    "type": "NORMAL",
    "status": "PENDING",
    "notes": "string"
  }
]
```

GET /api/v1/maintenance/{id}

Description:

Return the maintenance requests by id.

Response 204:

```
{
  "id": 0,
  "machineId": 0,
  "technicianId": 0,
  "type": "NORMAL",
  "status": "PENDING",
  "notes": "string"
}
```

GET /api/v1/maintenance/technician/{technicianId}

Description:

Return all maintenance requests signed to the technician.

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "technicianId": 0,
    "type": "NORMAL",
    "status": "PENDING",
    "notes": "string"
  }
]
```

GET /api/v1/maintenance/machine/{machineId}

Description:

Return all maintenance requests signed to the machine.

Response 204:

```
[
  {
    "id": 0,
    "machineId": 0,
    "technicianId": 0,
    "type": "NORMAL",
    "status": "PENDING",
    "notes": "string"
  }
]
```

GET /api/v1/maintenance/stats/{techId}

Description:

Return the statics for the maintenance technician by Id.

Response 204:

```
{
  "tasksCompleted": 0,
  "tasksPending": 0,
  "averageRepairTime": 0,
  "assistedOthers": 0,
  "wasAssisted": 0,
  "available": true
}
```

GET /api/v1/maintenance/current/{technicianId}

Description:

Return the current state for the maintenance by technician id.

Response 204:

```
{
  "id": 0,
  "technicianId": 0,
  "technicianName": "string",
  "machineId": 0,
  "machineName": "string",
  "startTime": "2026-05-13T19:20:45.138Z",
  "endTime": "2026-05-13T19:20:45.138Z",
  "active": true
}
```

8.3 PUT

PUT /api/v1/maintenance/{id}

Description:

Update the maintenance state by id.

```
{
  "id": 0,
  "machineId": 0,
  "technicianId": 0,
  "type": "NORMAL",
  "status": "PENDING",
  "notes": "string"
}
```

Response 204:

```
{ ...maintenance updated... }
```

PUT /api/v1/maintenance/finish/{sessionId}

Description:

Update the maintenance state to completed by id.

Response 204:

```
OK
```

8.4 Delete

DELETE /api/v1/maintenance/{id}

Description:

Permanently deletes a maintenance request.

Response:

204 OK

9. Maintenance Log Controller

9.1 POST

POST /api/v1/maintenance/{maintenanceId}/log

Description:

Creates a log for the request maintenance.

```
{
  "title": "string",
  "description": "string",
  "hoursSpent": 0,
  "cost": 0,
  "partsUsed": "string",
  "technicianId": "string"
}
```

Response:

```
{
  "id": 0,
  "title": "string",
  "description": "string",
  "hoursSpent": 0,
  "cost": 0,
  "partsUsed": "string",
  "technicianName": "string",
  "createdAt": "string",
  "maintenanceId": 0,
  "machineId": 0
}
```

9.2 GET

POST /api/v1/maintenance/logs/all

Description:

Return all logs for all request maintenance.

Response:

```
[
  {
    "id": 0,
    "title": "string",
    "description": "string",
    "hoursSpent": 0,
    "cost": 0,
    "partsUsed": "string",
    "technicianName": "string",
    "createdAt": "string",
    "maintenanceId": 0,
    "machineId": 0
  }
]
```

GET /api/v1/maintenance/{maintenanceId}/logs

Description:

Return all logs for the request maintenance by ID.

Response:

```
[
  {
    "id": 0,
    "title": "string",
    "description": "string",
    "hoursSpent": 0,
    "cost": 0,
    "partsUsed": "string",
    "technicianName": "string",
    "createdAt": "string",
    "maintenanceId": 0,
    "machineId": 0
  }
]
```

10. Assistance Request Controller

10.1 POST

POST /api/v1/assistance-requests

Description:

Creates an assistance request.

```
{
  "machineId": 0,
  "problemId": 0,
  "requestedById": 0,
  "reason": "string"
}
```

Response:

```
{
  "id": 0,
  "problemId": 0,
  "problemDescription": "string",
  "machineId": 0,
  "machineName": "string",
  "requestedById": 0,
  "requestedByName": "string",
  "assignedTechnicianId": 0,
  "assignedTechnicianName": "string",
  "reason": "string",
  "status": "PENDING",
  "createdAt": "2026-05-13T19:35:22.956Z",
  "acceptedAt": "2026-05-13T19:35:22.956Z",
  "completedAt": "2026-05-13T19:35:22.956Z"
}
```

POST /api/v1/assistance-requests/{id}/assign

Description:

Assign a user to an assistance request.

Response:

```
{
  "id": 0,
  "problemId": 0,
  "problemDescription": "string",
  "machineId": 0,
  "machineName": "string",
  "requestedById": 0,
  "requestedByName": "string",
  "assignedTechnicianId": 0,
  "assignedTechnicianName": "string",
  "reason": "string",
  "status": "PENDING",
  "createdAt": "2026-05-13T19:38:44.349Z",
  "acceptedAt": "2026-05-13T19:38:44.349Z",
  "completedAt": "2026-05-13T19:38:44.349Z"
}
```

POST /api/v1/assistance-requests/{id}/complete

Description:

Assign a user to a complete assistance request.

Response:

```
{
  "id": 0,
  "problemId": 0,
  "problemDescription": "string",
  "machineId": 0,
  "machineName": "string",
  "requestedById": 0,
  "requestedByName": "string",
  "assignedTechnicianId": 0,
  "assignedTechnicianName": "string",
  "reason": "string",
  "status": "COMPLETE",
  "createdAt": "2026-05-13T19:41:16.632Z",
  "acceptedAt": "2026-05-13T19:41:16.632Z",
  "completedAt": "2026-05-13T19:41:16.632Z"
}
```

10.2 GET

GET /api/v1/assistance-requests

Description:

Return all assistance requests

Response:

```
[
  {
    "id": 0,
    "problemId": 0,
    "problemDescription": "string",
    "machineId": 0,
    "machineName": "string",
    "requestedById": 0,
    "requestedByName": "string",
    "assignedTechnicianId": 0,
    "assignedTechnicianName": "string",
    "reason": "string",
    "status": "PENDING",
    "createdAt": "2026-05-13T19:42:05.205Z",
    "acceptedAt": "2026-05-13T19:42:05.205Z",
    "completedAt": "2026-05-13T19:42:05.205Z"
  }
]
```

11. Technician Performance Controller

11.1 GET

GET /api/v1/technician

Description:

Return all technician

Response:

```
[
  {
    "id": 0,
    "name": "string"
  }
]
```

GET /api/v1/technician/performance

Description:

Return all technician performance.

Response:

```
[
  {
    "technicianId": 0,
    "name": "string",
    "assignedMachines": 0,
    "completedRepairs": 0,
    "avgRepairTime": 0
  }
]
```

12. Technician Performance Controller

11.2 GET

GET /api/v1/sensors/{machineId}

Description:

Return all machine's sensors by Id

Response:

```
[
  {
    "id": 0,
    "sensorType": "TEMPERATURE",
    "value": 0,
    "recordedAt": "2026-05-13T19:49:35.372Z",
    "machineId": 0
  }
]
```

GET /api/v1/sensors/{machineId}/history

Description:

Return all history machine's sensors by Id

Response:

```
[
  {
    "id": 0,
    "sensorType": "TEMPERATURE",
    "value": 0,
    "recordedAt": "2026-05-13T19:49:35.372Z",
    "machineId": 0
  }
]
```

13. Common Error Codes

200 OK	Successful operation
201 Created	Resource created
204 No Content	Successful deletion
400 Bad Request	Invalid input
401 Unauthorized	Missing/invalid token
403 Forbidden	Insufficient permissions
404 Not Found	Resource does not exist
409 Conflict	Duplicate or conflicting data
500 Internal Server Error	Unexpected server failure

14. Typical Usage Flow

1. User logs in and receives JWT
2. User reports a problem
3. System creates a maintenance task
4. Technician accepts the task
5. Technician completes the task
6. Problem is closed
7. Machine status updates automatically
8. Sensor readings continue monitoring machine health

15. Project/API location

GitHub:

G705

Project:

SmartSense is an industrial monitoring system that analyzes vibration, pressure, and temperature sensor data to enable predictive maintenance of machinery.

<https://github.com/detiuaveiro/ies2526-group-project-g705>

16. Credits

Project made by:

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2. Tiago Costa - 125943 - Product Owner
3. Pedro Rocha - 125296 - Architect
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